

following command:

```
# apt-get install gcc libncurses5-dev linux-headers-`uname
-r` lsscsi patch subversion
```

On RHEL/CentOS: Type the following to install on RHEL/CentOS:

```
# yum install gcc ncurses-devel kernel-devel lsscsi patch
subversion
```

1. Download SCST source code from the SVN repository using the following command:

```
# svn co https://svn.code.sf.net/p/scst/svn/trunk scst
```

2. Go to the SCST source directory. By default, the build mode is 'debug' which is meant for debugging SCST code. To change this to release mode, issue the following command:

```
# cd scst
# make 2release
```

3. Now, build and install SCST kernel modules using the following command:

```
# BUILD_2X_MODULE=y make all install
```

4. Verify that the installation is successful. Check if you have the required SCST kernel modules in place by using the following command:

```
# ls -l /lib/modules/`uname -r`/extra/scst*.ko
```

5. Insert SCST kernel modules using the *modprobe* command shown below:

```
# for j in scst scst_vdisk scst_user ; do modprobe $j ; done
```

6. Create a virtual device, which will be exported as LUN by SCST using iSCSI, as follows:

```
# dd if=/dev/zero of=/mnt/disk01 bs=1024k count=512
```

7. Create a sample configuration file, which will export this newly created device as LUN, using the following command:

```
# cat << /etc/scst.conf
HANDLER vdisk_fileio {
    DEVICE disk1 {
        filename /mnt/disk1
    }
}
```



Figure 1: Network topology of iSCSI infrastructure

```
TARGET_DRIVER iscsi{
    TARGET iqn.2015-11.com.example:storage1 {
        enabled 1
        LUN 0 disk1
    }
}
EOF
```

8. Restart SCST using the SCST service, as follows:

```
# /etc/init.d/scst restart
```

On initiator systems

Each operating system comes with the iSCSI initiator package. Otherwise, you can install the iSCSI initiator package using the following commands.

On RHEL/CentOS: Give the following command to install the initiator package:

```
# yum install -y iscsi-initiator-utils
```

On Debian: To install the initiator package on Debian, type:

```
# aptitude install open-iscsi
```

1. Discover the target using the following command. Make sure that the target and initiator have connectivity between them.

```
# iscsiadm -m discovery -t sendtargets -p <IP_ADDRESS_TARGET>
```

Notice that you will target *iqn.2015-11.com.example:storage1*, which is exported by the iSCSI target.

2. Do log in using the following command. Replace *targetname* with the above iSCSI target name.

```
# iscsiadm -m node -targetname <targetname> -p <IP_ADDRESS_TARGET> --login
```

3. Check that LUN is visible in the initiator environment using *fdisk*:

```
# fdisk -l
```